Monthly Intelligence Report

Edition 003 — The Atacama–Central Transmission Corridor Corridor

Where demand surges are outrunning transmission capacity. edition · June 2026 · SSI v4.0.2

CONTENTS

SECTION A

Executive Summary

SUBSTATIONS SCORED

1,095

Major + distribution across SEN (Sistema Eléctrico Nacional)

GRID LINES MAPPED

~2,081

~38,784 km transmission & distribution

MC SIMULATIONS

85.0 M

10,000 per substation

PUBLIC DATA SOURCES

20+

95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions. Its engine processes 146,000+ source data points per run, executes 85.0 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 1.78 billion arithmetic operations — producing 469,404 output data points with full uncertainty quantification across 1,095 substations and ~2,081 grid lines spanning ~38,784 km.

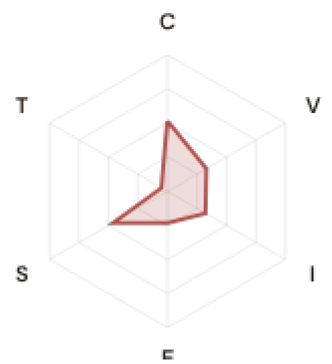
Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 20+ verified public sources (CEN, CNE, INE, DMC, SERNAGEOMIN, CSN, ACERA), making the methodology fully auditable and reproducible. Initially covering Chile's SEN (SIC and SING) and isolated systems, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Subestación Paso Hondo

Ñuble, Ñuble · 66 kV

0.471	Medium	0.153	54th
R_FINAL	CLASSIFICATION	CI WIDTH	FLEET PERCENTILE



COMPONENTS

C Continuity: 0.512 / 0.30 (171%)

I Infrastructure: 0.321 / 0.25 (128%)

S Saturation: 0.461 / 0.20 (231%)

V Voltage: 0.329 / 0.10 (329%)

E Economic: 0.230 / 0.10 (230%)

T Transition: 0.050 / 0.05 (100%)

MODIFIERS

R3 Consequence: 1.049 ↑ amplifies

R4 Graph Criticality: 0.925 ↓ dampens

R6a Restoration: 1.011 ↑ amplifies

R6b Seismic: 1.226 ↑ amplifies

R7 Digital Readiness: 1.018 ↑ amplifies

This month's spotlight — automatically selected for June 2026 — is **Subestación Paso Hondo** in Ñuble, Ñuble. With an R_final of 0.471 (Medium band, 54th fleet percentile), its risk profile is dominated by the V (Voltage) component at 329% of its weight ceiling, followed by S (Saturation) at 231%. Modifiers collectively shift R_base by 22.4%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Chile's Energy Security Board and network operator strategic planning requires.

CYBER HIGH EXPOSURE
0
0.0% of fleet

BLIND SPOTS
49
High/Critical + R7 < median

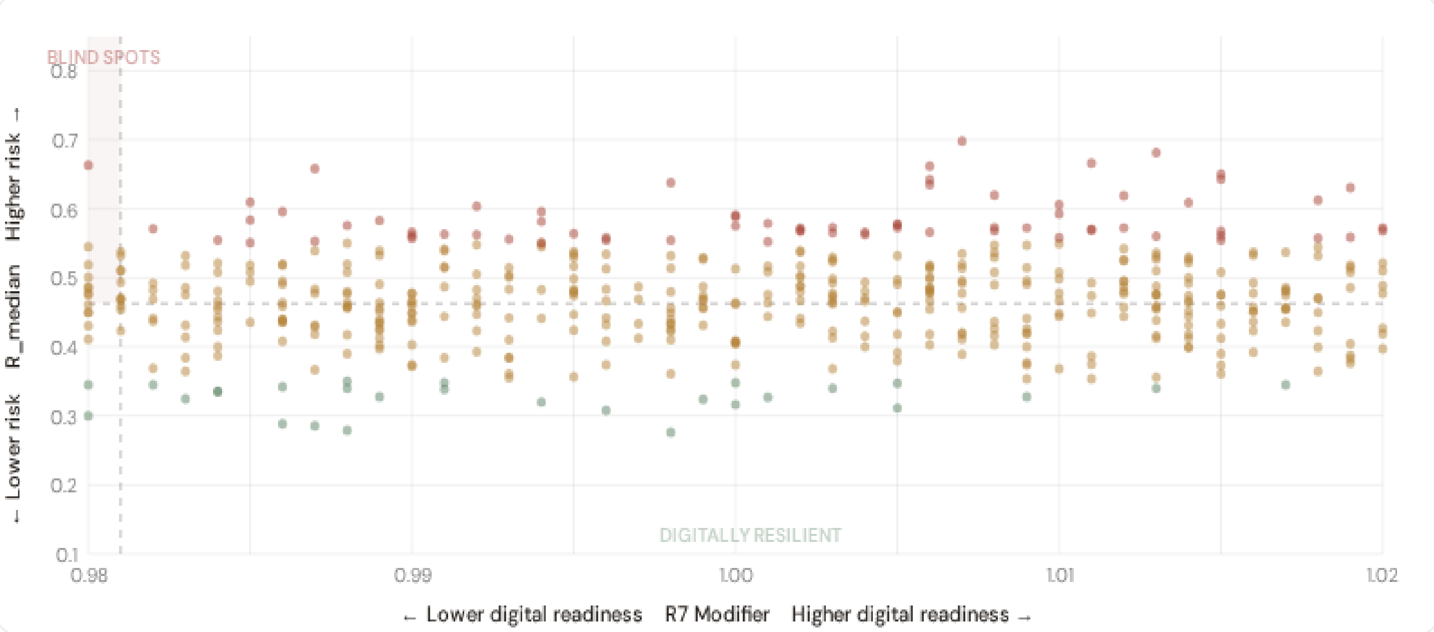
AVG R7 MODIFIER
0.9802
Range [0.99, 1.05]

HIGH-CONSEQUENCE SUBS
275

25.1% · R3 ≥ 1.04

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical Dashed lines = fleet medians

The cyber-physical matrix identifies **49 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness (R7 < 0.9810). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Antofagasta (20), Biobío (15), O'Higgins (4)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

🏭 **Capital-Intensive / High-Consequence** Est. VoLL: CL$12,000–24,000/kWh (CLP)

Capital nodes — highest economic loss per interruption (semiconductor fabs, data centers, refineries, hospitals)

275 substations (25.1%) High/Critical: **67** (24.4%) Avg R: **0.494** Avg E: **0.371** / 0.10

Low 13 Med 195 High 67 Crit 0

🔧 Industrial / Medium-Large Enterprise Est. VoLL: CL$6,000–12,000/kWh (CLP)

Industrial belt — significant continuity requirements (manufacturing, logistics hubs)

273 substations (24.9%) High/Critical: **32** (11.7%) Avg R: **0.470** Avg E: **0.374** / 0.10

Low 8 Med 233 High 32 Crit 0

🏪 Commercial / SME-Dense Est. VoLL: CL$2,400–6,000/kWh (CLP)

Service economy + retail — moderate individual impact (commercial districts, mixed-use)

281 substations (25.7%) High/Critical: **25** (8.9%) Avg R: **0.454** Avg E: **0.377** / 0.10

Low 23 Med 233 High 25 Crit 0

🌾 Light / Agricultural / Rural Est. VoLL: CL$800–2,400/kWh (CLP)

Sparse activity, agricultural land — lower VoLL (residential, agricultural)

266 substations (24.3%) High/Critical: **15** (5.6%) Avg R: **0.432** Avg E: **0.364** / 0.10

Low 43 Med 208 High 15 Crit 0

Value of Lost Load (VoLL) context: ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **67 substations** combine capital-intensive economic fabric (R3 ≥ 1.04) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Antofagasta (49), O'Higgins (33), Metropolitana (30)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 12.3%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 State & Comuna Digital Readiness Ranking

comunas ranked by average R7 modifier — lower values indicate weaker digital infrastructure (broadband penetration, smart meter coverage). Cross-referenced with average R_median to identify regions combining digital exposure with high grid stress.

WORST DIGITAL READINESS — TOP 15 REGION REGIONS

Longer bar = higher R7 = better digital infrastructure.

1	Los Ríos	0.9695
2	Magallanes	0.9699
3	Valparaíso ⚠ high R	0.9734
4	Tarapacá ⚠ high R	0.9753
5	Araucanía	0.9784
6	Aysén	0.9787
7	Arica y Parinacota ⚠ high R	0.9800
8	Metropolitana	0.9800
9	Biobío ⚠ high R	0.9804
10	Antofagasta ⚠ high R	0.9805
11	Coquimbo	0.9817
12	Los Lagos	0.9817
13	Ñuble ⚠ high R	0.9818
14	Maule ⚠ high R	0.9823
15	Atacama	0.9825

region-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Los Ríos** has the lowest average R7 (0.9695), while **O'Higgins** leads at 0.9844 — a spread of 0.0149 across the R7 range. **5 region regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (June 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 0.9802, median = 0.9810; 0 substations classified HIGH cyber-exposure; 49 blind-spot substations (High/Critical risk + below-median R7); 0 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **— verified public data sources** feeding — variables across — components and — modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES

—

— variables

WEEKLY / MONTHLY

0

High-frequency feeds

QUARTERLY / ANNUAL

0

Regular refresh cycle

STATIC / DERIVED

0

Standards + scenarios

Data Source Registry — June 2026

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

C.2 Fleet Score Snapshot

FLEET SIZE

1,095

substations scored

MEDIAN R

0.463

P5=0.333 · P95=0.579

HIGH + CRITICAL

139

12.7% of fleet

HIGH CONFIDENCE

96%

1055 subs · CI/R < 0.50

BAND DISTRIBUTION

Low 87 (7.9%) Medium 869 (79.4%) High 139 (12.7%) Critical 0 (0.0%)

No primary data sources have been updated since the v4.0.2 launch baseline. All **1,095 substation scores remain unchanged** this edition. The fleet median R of 0.463 (mean 0.463) reflects a distribution skewed toward the lower bands, with 7.9% in Low and 79.4% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

0 changes tracked across P0, P1, and P2 releases

SECTION D — MONTHLY DEEP-DIVE

The Atacama–Central Transmission Corridor Corridor: Where Demand Surges Outrun Transmission Capacity

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** Atacama–Central Transmission Corridor corridor · **002** South Chilean DER Saturation · **003** Atacama Extreme Heat Infrastructure Stress · **004** Metropolitan-Remote economic impact disparity · **005** Energy transition risk (Mining regions) · **006** Infrastructure age vs smart meter rollout · **007** Stress–Resilience quadrant trends · **008** T-component forward projections · **009** Comuna Grid Health Cards · **010** Academic validation and peer feedback · **011** Latin American replication feasibility · **012** Year in review.

D.1 Why the Atacama–Santiago Corridor, Why Now

ATACAMA–CENTRAL SUBSTATIONS

118

65 EHV · 53 HV

HIGH BAND

34

28.8% of corridor

CORRIDOR MEDIAN R

0.512

Fleet median: 0.463

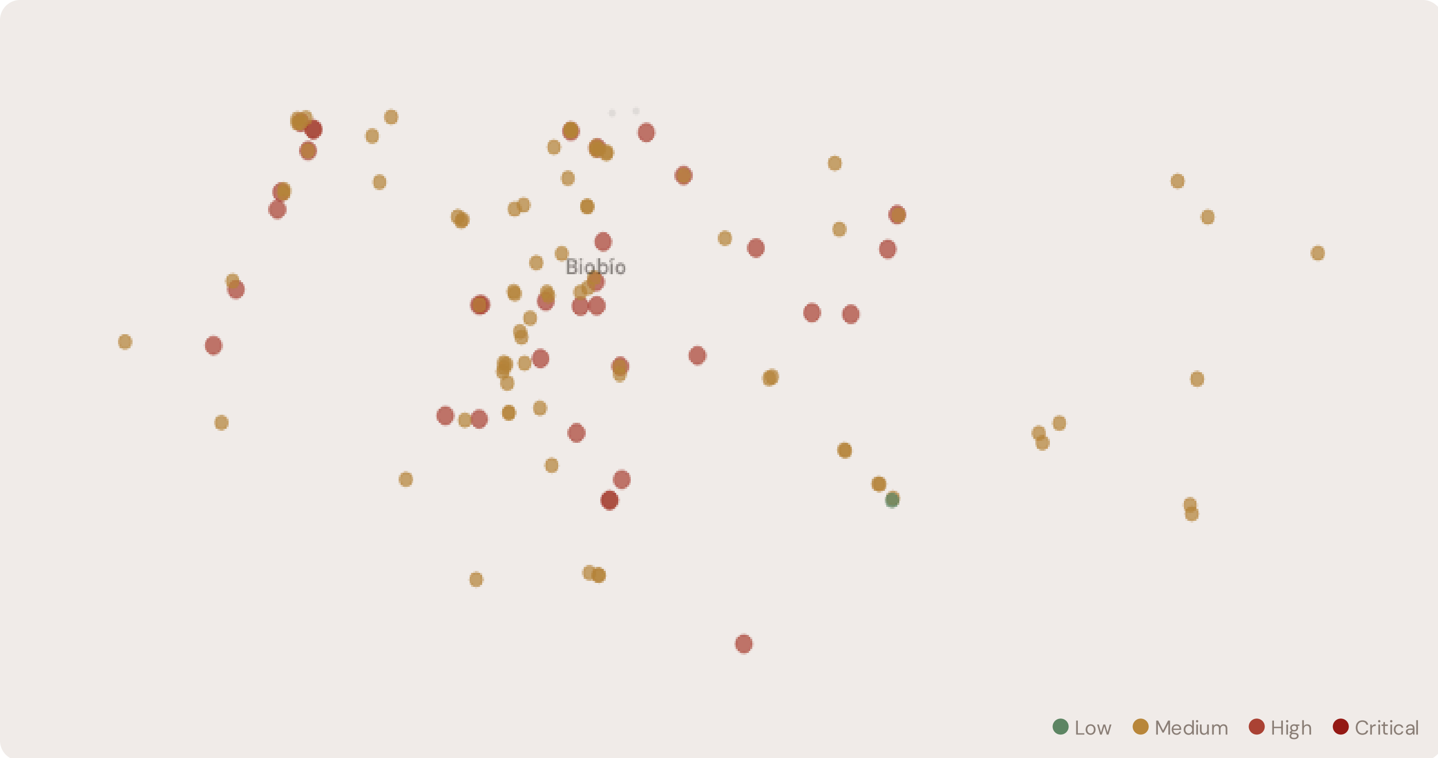
WORST COMUNA REGION

Biobío

Avg R: 0.516 · 118 substations

The Biobío hosts **118 substations** across 1 region regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **34 substations (28.8%)** are classified High, with only **1** in the Low band. 76% of corridor substations score above the national fleet median of 0.463. The corridor median R of 0.512 reflects the local risk profile. The corridor concentrates the country's industrial heartland. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under national digitalisation timelines.

D.2 Corridor Map and Scoring



SUBSTATION	COMUNA REGION	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Subestación SF Energía	Biobío	0.681	High	0.566	0.351	0.334	0.371	0.777	0.063	S
Subestación Angol	Biobío	0.676	High	0.521	0.430	0.357	0.369	0.624	0.097	
Subestación Lota	Biobío	0.666	High	0.387	0.374	0.296	0.293	0.701	0.074	S
Subestación Central Abanico	Biobío	0.663	High	0.488	0.490	0.267	0.400	0.601	0.099	
Subestación Central Itata	Biobío	0.648	High	0.584	0.488	0.317	0.318	0.529	0.075	
Subestación BIO 533	Biobío	0.643	High	0.546	0.356	0.387	0.392	0.558	0.085	
Subestación Papelera Río Vergara	Biobío	0.640	High	0.521	0.481	0.341	0.340	0.685	0.074	S
Subestación BIO 756	Biobío	0.619	High	0.538	0.424	0.383	0.375	0.689	0.104	S
Subestación Duqueco	Biobío	0.610	High	0.561	0.360	0.280	0.404	0.567	0.065	

SUBSTATION	COMUNA REGION	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Subestación BIO 548	Biobío	0.608	High	0.607	0.395	0.340	0.352	0.558	0.097	

… 108 additional substations — [explore full corridor in Digital Twin →]

D.3 Component Analysis

COMPONENT AVERAGES — BIOBÍO VS FLEET

C — Continuity 0.444 vs 0.448 (-1%)

I — Infrastructure 0.297 vs 0.329 (-10%)

S — Saturation 0.653 vs 0.474 (+38%)

V — Voltage 0.351 vs 0.350 (+1%)

E — Economic 0.355 vs 0.372 (-4%)

T — Transition 0.081 vs 0.197 (-59%)

MODIFIER AVERAGES — BIOBÍO VS FLEET

R3 Consequence	1.001	fleet 0.998	→
R4 Graph Crit.	0.983	fleet 0.980	↓
R6a Restoration	1.020	fleet 1.021	↑
R6b Seismic	1.270	fleet 1.200	↑
R7 Digital Read.	0.980	fleet 0.980	↓

The dominant risk driver across the Biobío corridor is **E (Economic)**, averaging 0.355 against a fleet average of 0.372 — occupying 355% of its weight ceiling (w = 0.10). The second contributor is **V (Voltage)** at 0.351 vs fleet 0.350 (351% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.001 (fleet: 0.998), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 1.270, capturing the local seismic exposure — a dimension invisible to every competing framework.

D.4 Comuna Region Breakdown

REGION RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

Biobío 118 subs · avg R = 0.516 · H:34 M:83

The region regions-level breakdown reveals significant intra-regional variation. **Biobío** leads with a mean R of 0.516 across 118 substations, while **Biobío** scores 0.516 — a spread of 0.000 within a single corridor. The SSI's substation-level granularity reveals that the Biobío is not uniformly stressed but contains distinct sub-corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

4 Biobío substations score R ≥ 0.650, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the Biobío sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

SECTION E — RECURRING MODULE

Latin American Context Card

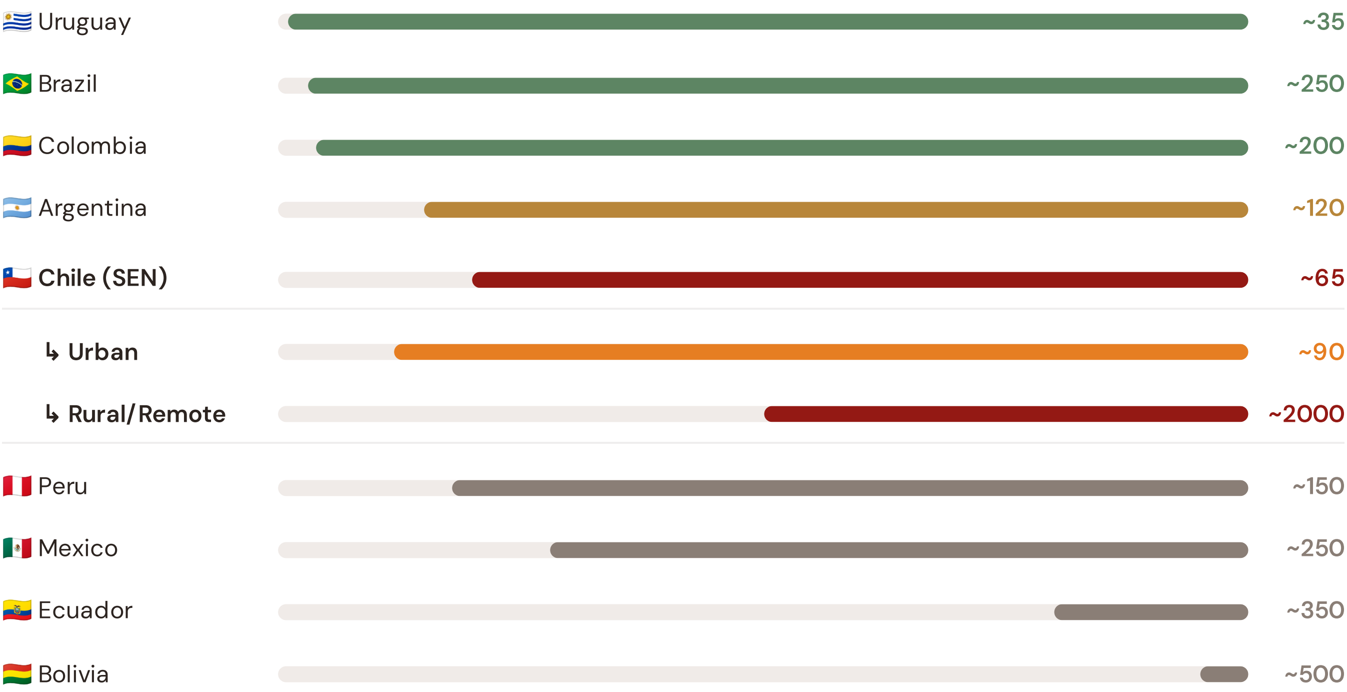
How this works: Each edition includes one Latin American Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Atacama–Santiago corridor analysis hinges on Chile's continuity performance relative to regional peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Japan (003), CMIP6 climate hazard vs SE Asian peers (004), etc. The underlying data updates annually with CEN/CNE and national regulator publications.

LATIN AMERICAN CONTEXT CARD — SAIDI BENCHMARKING

Ed. 001 · June 2026

Unplanned SAIDI (min/customer/year) — Selected Latin American Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



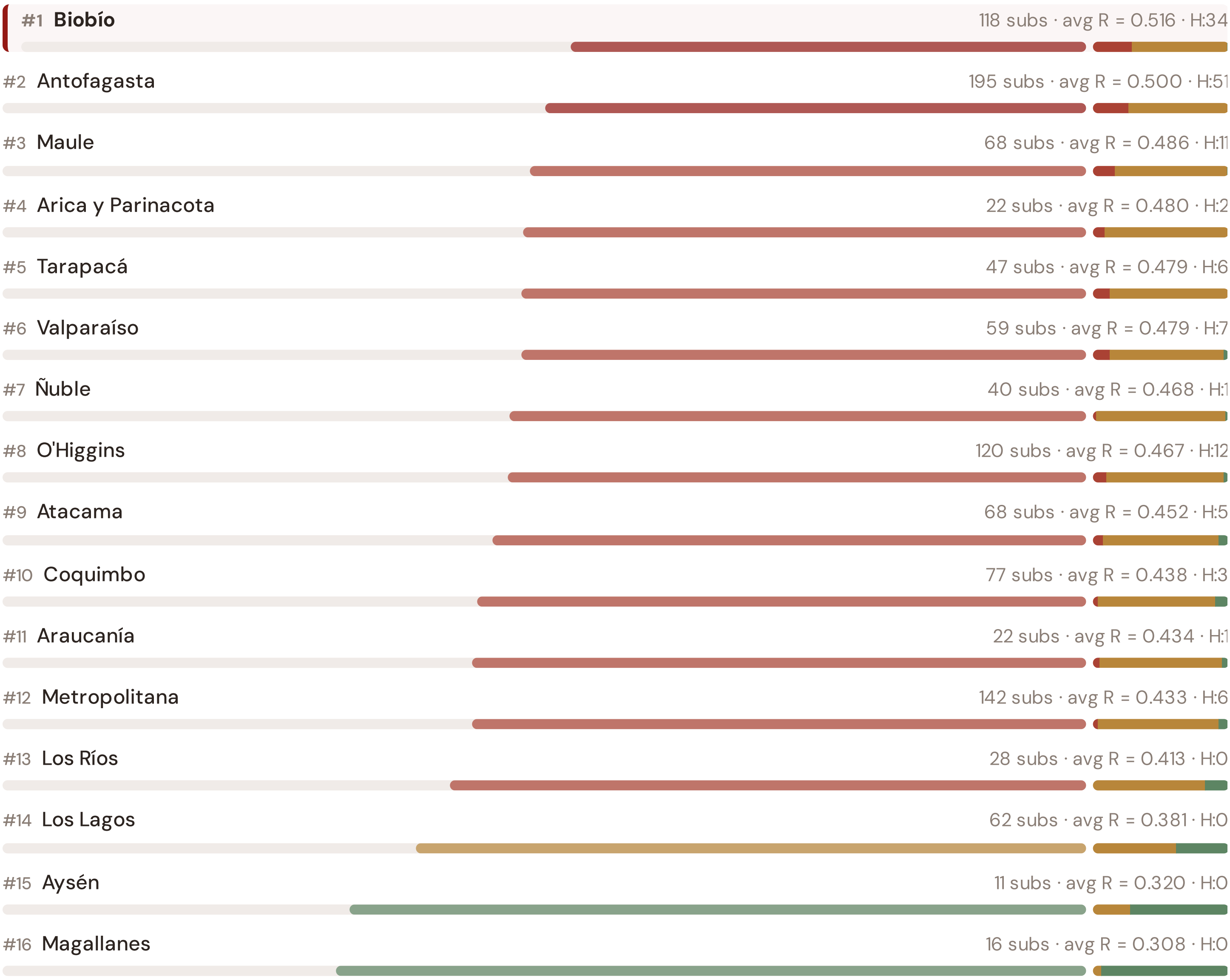
Source: CEN Reliability Panel Annual Report (2024); CNE State of the Energy Market (2025); World Bank ESMAP (2023); individual national regulator publications. Chile sub-categories from CNE urban/rural/remote classifications. · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually.

This month's key insight: The Biobío corridor deep-dive shows **118 substations** (of 118) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.444 is **1.0× the fleet average** of 0.448. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

State/Territory Risk Ranking — All 16 Chilean Regiones & Territories

Where does the Atacama–Central corridor sit within the national landscape? States sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.



The Biobío ranks **#1 of 16 region regions** by mean R_median, placing it among the upper tier of national risk. The three most stressed are Biobío, Antofagasta, Maule, while the three most resilient are Magallanes, Aysén, Los Lagos. 8 of 16 score above the fleet average of 0.463. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the SEC/CEN and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

Layer 1 — Data	Layer 2 — Valuation	Layer 3 — Neural Net	Layer 4 — Coverage	Layer 5 — Platform
SSI v4.0 component taxonomy, normalisation, MC engine	DSO/TSO + Community value stacks, TCO, real options	GBT+DNN ensemble, transfer learning, SHAP	Country prioritisation, DCI, registry, calibration	Products, API, revenue model, unit economics

LAYER 3 §0.5.6 · June 2026

Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history. Transparency converts model outputs into investment decisions.

KEY CONCEPTS

- → SHAP: game-theoretic feature attribution per prediction
- → Waterfall: fleet average → substation prediction
- → Direct traceability to SSI components and data sources
- → Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The training set comprises 1,095 substations with complete feature vectors. Average CI_width of 0.136 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 87 Low, 869 Medium, 139 High, 0 Critical — ensures balanced training across all risk bands.

Coming next month: LAYER 3 Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

Looking Ahead

NEXT MONTH — EDITION 02

Mountain Region

Deep-dive into Region 10's grid risk profile — substation map, component analysis, region regions breakdown. European Context Card: Light-rural tier — limited substation density.

NEXT DATA REFRESH

Q3 2026 — 0 Quarterly Sources

none are due for refresh in Q3. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **TBD** (? variables →).

FLEET SPOTLIGHT

Biobío: Highest Risk Concentration

Biobío has the highest concentration of High/Critical substations: **34 of 118 (29%)** are in the upper risk bands — avg R = 0.516. This makes it the most acute risk corridor in the fleet.

Edition 02 will be published on the second Thursday of July 2026. Deep-dive region: **Region 10**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent an CEN participant, distribution company, transmission operator, regional government, research institution, or industry body and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 003

Published 11 June 2026 · SSI Index v4.0.2 · 1,095 substations · ~2,081 grid lines · 6 components · 20 metrics · 6 modifiers

10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5